

Description of data processing to reproduce the observational results in Fan et al. 2018 (<https://doi.org/10.1126/science.aan8461>)

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Here we describe the data processing details for reproducing the results from the observational part of Fan18. The data in Fan18 was analyzed by Yuwei Zhang who was a Ph.D. student in late 2016, and the record was lost due to the job change and computer changes over the years. ORV23 questioned the unclear or missing details in the short paper of Fan18. We reanalyzed the 17 observational cases based on the information available in Fan18 and shared the data processing details and code here. Our reprocessed results, along with significance tests, support the original conclusions of Fan18.

Step 1. Selection of 17 local convective system cases. We first selected the locally occurring system (LOS) cases identified as occurring in the afternoon and characterized as scattered convections generated through solar heating at the surface based on Tang et al. (2016). Tang et al. (2016) identified cases from the surface radar and satellite images and did not describe the specific criteria used. While working for Fan18, we communicated with Dr. Shuaiqi Tang and documented the criteria used for identifying these cases: (1) deep convection occurred between 11:00-19:00 local time and (2) no deep convection occurred during the 3 hours preceding the locally occurring systems. The second criterion excludes cases that have previous deep convective clouds in the morning, which are likely from the propagating systems. Note the criteria here for selected cases and the identification of the specific convective period of each case is detailed in Step 2. The deep convective clouds are defined by the maximum echo height determined by reflectivity > 0 dBZ higher than 10 km. To look at aerosol-cloud interaction, we also excluded cases having bad aerosol data based on the measurements from the Scanning-Mobility Particle Sizer (SMPS; available from the DOE ARM Data Archive <https://adc.arm.gov/discovery>). The 17 selected cases are shown in Table 1.

Step 2. Identification of the convection period and the analysis of the RWP data. The convection initiation was defined based on the RWP data and the first appearance of $Ze > 0$ dBZ above 4 km (i.e., the start of mixed-phase clouds). Note that the convection initiation occurs before the deep convection initiation and as a result some cases begin before 11:00 local time. The convection end time was defined as the time when there is no $Ze > 0$ dBZ between 1.5-4 km altitudes. From the convection initiation to the end of the convection (shown in Table 1 for all 17 cases), we only chose the data with echo classification defined as “convection” or “weak convection” in the RWP dataset. Because of this further selection, the convection period in our approach does not need to be so strict since other clouds such as shallow and stratiform can be filtered out by this criterion. However, the convection initiation time is important to the identification of aerosol sampling time (see step 3 below). For each case, we calculated the profile of the mean of top 10th percentiles of updrafts > 0 m/s over the identified convection period. We excluded the times with updraft data outside of the range from -40 to $+40$ m s⁻¹ from the analysis based on the recommendations of the data developer, Dr. Scott Giangrande. For the 0423 case, the values in the PBL before 1860 UTC mostly exceed 30 m s⁻¹ and thus are removed. For the plots showing results with different aerosol groups, the vertical profile of the top 10th percentile updrafts is calculated first, then averaged over the cases in every aerosol group. The

moving average technique is applied to smooth values vertically, with 1 km for altitudes below 4 km and 2 km for above altitudes to show a neater plot, as the RWP data are spiky.

Step 3. Identification of aerosol sampling period and analysis of aerosol data. For each case, we selected a rough 30-min aerosol sampling period. The aerosol data are from SMPS available from the ARM archive. The 30-min sampling window is within the 2-h period ahead of the convection initiation time. We need the environment aerosol concentration with the least cloud and rain interference for this analysis; thus, the 30-min aims to lie in cloud/rain-free conditions if possible, to capture the high aerosol concentrations before the convective storms. If it is cloudy all the time over the 2-hr period, we selected a time window with the highest aerosol concentrations since we believe high values of aerosols would be closer to the clear-sky aerosol conditions or more representative of aerosol sources in general. Therefore, the 30-min time window is manually identified for each case based on this scientific rationale. The aerosol sample periods for all 17 cases are provided in Table 1.

After the 30-min aerosol sampling window was identified, the mean aerosol number concentration was obtained for each case. The four aerosol groups were obtained based on the mean aerosol concentrations of the 17 cases. The mean profile of top 10th percentile updraft velocity is calculated for each case. For the plot showing results from each aerosol group, the mean profile of each aerosol group is obtained by averaging of the mean profiles of all cases in the group.

Table 1 Sampling time periods for aerosol and updraft velocity (UTC is 4 hours earlier than the local time)

Date of Case	Aerosol period (UTC)	Convection period (UTC)
20140311	1332-1401	1440-1852
20140317	1637-1706	1728-1906
20140321	1902-1931	1935-2028
20140322	1332-1401	1508-1654
20140323	1642-1711	1712-1930
20140326	1617-1646	1817-2235
20140401	1109-1138	1309-1905
20140412	1533-1602	1642-1807
20140418	1916-1945	2021-2136
20140420	1751-1820	1824-2015
20140421	1602-1631	1633-1802
20140423	1712-1741	1803-1910
20140516	1615-1644	1645-1815
20140519	1520-1549	1550-1835
20140520	1545-1614	1618-1742
20140530	1333-1402	1455-1809
20140531	1239-1308	1401-1624

The NCL code for processing the data is included in the file “code.ncl” of this repository. The input for running the code is “input_time.txt”.

Reprocessed results:

Figure 1b shows the reprocessed results. First, the increase in mean w from the low to high aerosol groups in the original plot of Fan18 (Fig. 1a) is reproduced. There are small quantitative differences in the mean because minor details in processing the data for Fan18 could not be recalled. We now conducted the bootstrapping significant tests for the increased w between the cleanest and most polluted groups using the original data of the cases in each aerosol group following the procedures provided by ORV, and the increases in w pass the significance test with a p-value of 0.017 for the entire vertical profile and a very low p-value of 0.007 for the altitudes > 5.5 km.

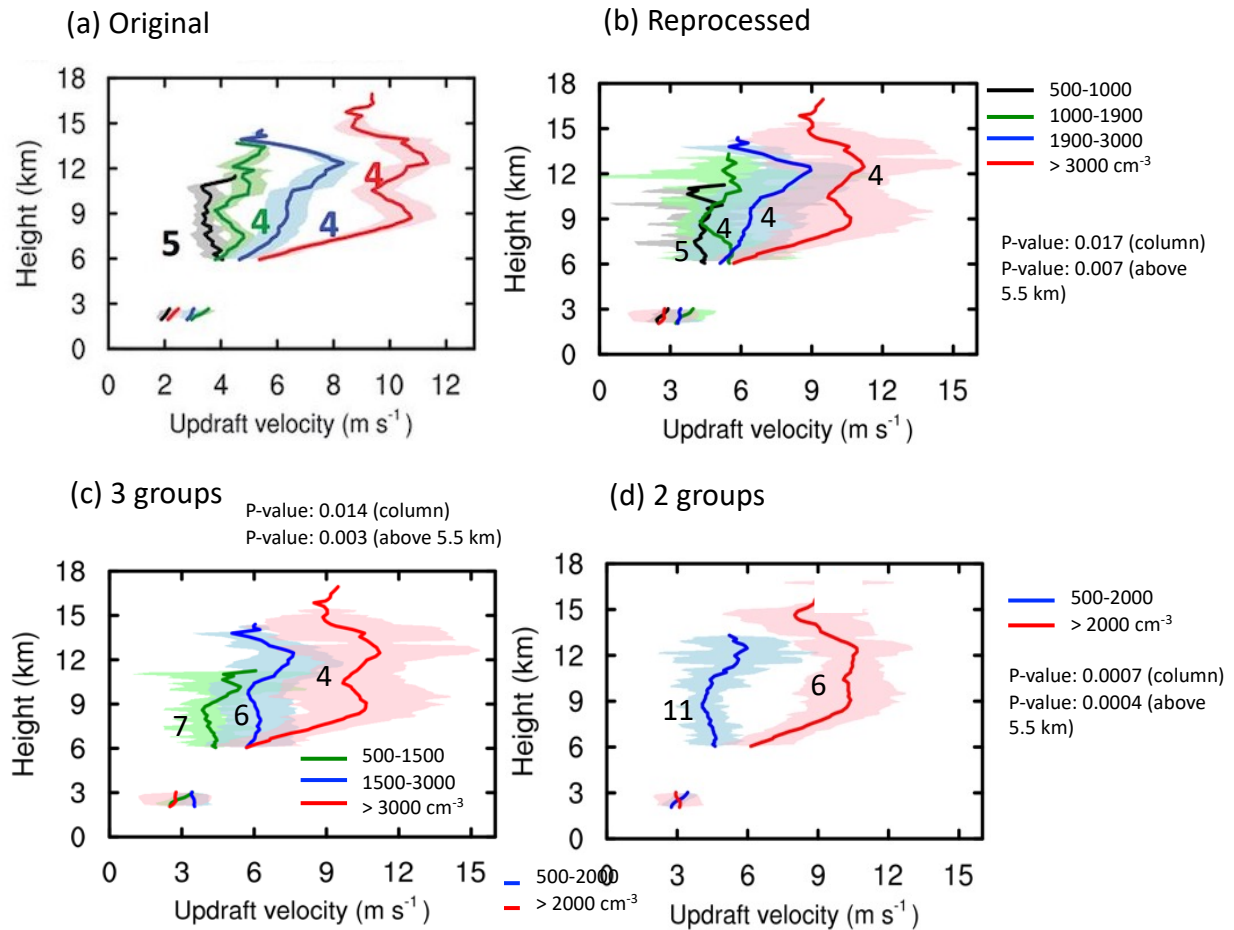


Fig.1 Vertical profiles of the updraft velocity averaged over the top 10 percentiles of updrafts for the four aerosol groups from lower to higher N_a for $D > 15$ nm: (a) the original plot in Fan18 and (b) the reproduced plot. The numbers on the lines of (a) denote the number of cases for each aerosol group which remains the same for (b).

Notably, the standard error in Fig. 1b is larger compared to the original result of Fan18 (Fig. 1a), which is due to a bug in calculating the standard error. The standard error was calculated with two steps in our code and the denominator should be the case number of a specific aerosol group in the second step. The sample size of each case that was used in our code was shared with ORV in the comment-reply process. They pointed out the bug and the correct calculation showed a much larger standard error (Fig. 1b). However, the larger standard error does not affect the conclusion, as the systematic differences between clean and polluted conditions passed the significant tests, which were carried out with the original data of each case.

The choice of binning into 4 groups in Fan18 particularly concerning the bin boundary of 1900 cm^{-3} to separate the 3rd and 4th aerosol groups might be questioned. We want to clarify that this choice has a consideration of the balance of cases between the aerosol groups (to avoid 2-3 cases in the most polluted group). Using different bin boundaries can lead to the clumping of some groups. However, the clumping of the neighboring groups does not affect the conclusion (e.g., the clumping of the black and green curves in Fig. 1b still represents a low aerosol group and the clumping of blue and red represents the high aerosol group, and the conclusion of increasing w with aerosols from clean to polluted cases is not affected. The 4-group binning approach shown in Fan18 might not be perfect, and we could have used 3 or even 2 aerosol bins to show the trend more clearly. Here we showed the results by binning 17 cases into 2 and 3 groups (Fig. 1c and 1d). The increasing trend is more significant with 3 aerosol groups than with the 4 groups (Fig. 1c), because the bootstrapping significance test is passed with a very small p-value (0.014 for the vertical profile and 0.003 for altitudes above 5.5 km). With only 2 aerosol groups, the increase in w is very large and there is no overlap in standard errors (p-value of nearly zero; Fig. 1d)